

3-Mark Questions & Answers

Q1. Explain the functional units of a computer system with a diagram.

A computer system consists of five functional units:

1. Input Unit: Accepts data and instructions from the user/environment (e.g., Keyboard, Mouse).

2. Mem

text

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Q2. Differentiate between Computer Architecture and Computer Organization (any 3 points).

| Feature | Computer Architecture | Computer Organization |

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Q3. Explain the instruction cycle (Fetch-Decode-Execute-Store) briefly.

The Instruction Cycle is the continuous process by which a CPU processes machine instructions:

1. Fetch: The Control Unit reads the instruction byte from main memory using the address in the Program Counter (PC / IP).

2. Deco

Q4. List and explain any three performance measures of computer architecture.

1. CPU Execution Time: The total time the processor spends actively computing a task. Lower execution time means better performance.

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Q5. Explain register indirect and indexed addressing modes with examples.

1. Register Indirect Addressing Mode:

* The

2. Indexed Addressing Mode:

* The

Q6. Explain the memory organization (segmented) of the 8086 with the physical address formula.

The 8086 features a 20-bit address bus, allowing it to address 1 MB (2^{20} bytes) of memory. Since internal registers are only 16-bit, memory is organized into logical 64 KB segments:

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Physical Address Calculation Formula:

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Q7. What are the advantages of the Von Neumann model? (any 3)

1. Simplicity and Low Cost: Uses a single shared memory and bus system for both code and data, simplifying hardware design and reducing manufacturing costs.

2. Flexi
code as

Q8. Explain any three arithmetic instructions with syntax and example.

1. ADD (Addition): Adds source to destination and stores result in destination.

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Q9. Explain PUSH and POP instructions with examples.

1. PUSH Instruction:

* Push

2. POP Instruction:

* Pops

Q10. Differentiate between JMP and CALL instructions.

| Feature | JMP (Jump) Instruction | CALL Instruction |

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Q11. Explain SHL, SHR and SAR instructions with examples.

1. SHL (Shift Logical Left): Shifts bits left by count, filling empty LSBs with 0. MSB goes into Carry Flag (CF). Equivalent to multiplying by 2.

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Q12. Explain the relationship: Clock Rate, Clock Cycle Time, and CPU Time.

* **Clock Cycle Time (**T_c**):** The duration of one clock tick (e.g., **0.5 ns**).

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Equation:

Clock

* **CPU Time:** The product of total clock cycles and cycle time:

Equation:

CPU Ti

* *Impact:* Increasing Clock Rate reduces Clock Cycle Time, directly reducing CPU Time and improving speed.
